

Most Asked React Interview Questions With Solutions & Explanations

1. What is Virtual DOM?

Answer:
Virtual DOM is a lightweight copy of real DOM.

Explanation:
React compares old Virtual DOM with new Virtual DOM.
Then only changed elements update in real DOM.

Benefits:
- Faster rendering
- Better performance

2. Difference Between State and Props

Props:
- Passed from parent
- Read only

State:
- Managed inside component
- Can change

Example:
`const [count, setCount] = useState(0);`

3. What are Hooks?

Hooks allow functional components to use React features.

Common Hooks:
- `useState`
- `useEffect`
- `useRef`
- `useMemo`
- `useCallback`

4. Difference Between `useEffect` and `useLayoutEffect`

`useEffect`:
Runs after browser paint.

`useLayoutEffect`:
Runs before browser paint.

Explanation:
`useLayoutEffect` blocks painting.

5. What is Prop Drilling?

Passing props through multiple nested components.

Problem:
Unnecessary prop passing.

Solution:
Use Context API or Redux.

6. What is Context API?

Used for global state sharing.

Example:

```
const UserContext = createContext();
```

Benefits:

Avoids prop drilling.

7. Difference Between Controlled and Uncontrolled Components

Controlled:

Uses React state.

Uncontrolled:

Uses DOM refs.

Example:

```
<input value={name} onChange={handleChange} />
```

8. What is React.memo?

Prevents unnecessary re-renders.

Example:

```
export default React.memo(Component);
```

Used for performance optimization.

9. Difference Between useMemo and useCallback

useMemo:

Memoizes values.

useCallback:

Memoizes functions.

Used to optimize performance.

10. What are Custom Hooks?

Reusable hook logic.

Example:

```
function useFetch() {}
```

Benefits:

Code reusability.

11. What is Reconciliation?

React compares Virtual DOM changes.

Only changed parts update in real DOM.

12. Why Keys are Important in React?

Keys help React identify changed elements.

Example:

```
users.map(user => (  
  <li key={user.id}>{user.name}</li>  
))
```

13. What Causes Re-renders in React?

Causes:

- State updates
- Props changes
- Parent re-render

Optimization:

React.memo
useMemo
useCallback

14. How React Lifecycle Works?

Three phases:

- Mounting
- Updating
- Unmounting

useEffect handles lifecycle behavior.

15. What is Lazy Loading?

Loads component only when needed.

Example:

```
const Home = React.lazy(() => import('./Home'));
```

16. What is Code Splitting?

Breaks large bundle into smaller chunks.

Benefits:

- Faster loading
- Better performance

17. How to Optimize React Performance?

Techniques:

- React.memo
- useMemo
- useCallback
- Lazy loading
- Virtualization

18. Difference Between Redux and Context API

Redux:

Advanced state management.

Context API:

Simple global state.

Redux better for large apps.

19. What are Higher Order Components?

Function that takes component and returns component.

Example:

`withAuth(Component)`

20. What is Hydration in React?

Hydration attaches React events to server-rendered HTML.

Used in:

- Next.js
- SSR applications